

Task 3: Strategy Spotlight – The Covered Call

Objective: The objective of this project is to simulate and analyze the payoff profile of a Covered Call options strategy using Python.

Methodology:

Parameters are defined for hypothetical stock which is trading at \$100. In the Covered Call, two legs used are

- Long Stock- Own 100 shares of the underlying asset.
- Short Call- Sell one Out-Of-The-Money (OTM) call option against those shares.

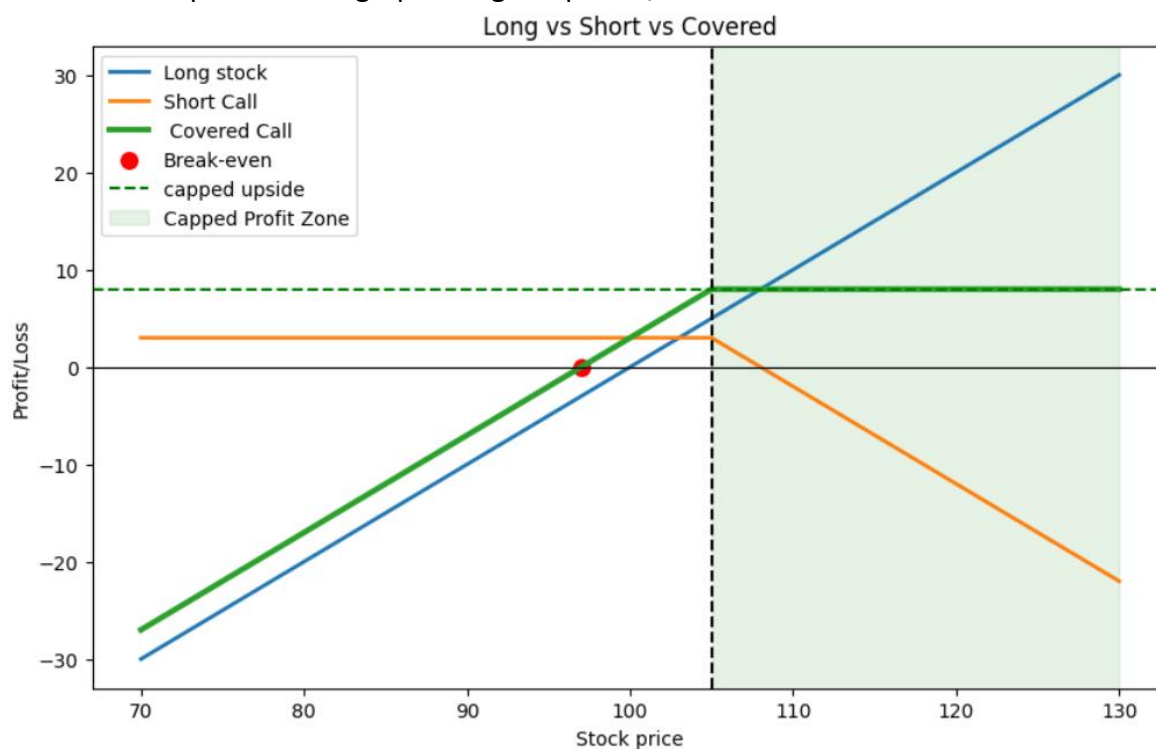
The strike price for the call option is set at \$105, premium is of \$3.00.

To simulate and analyse different scenarios, an array is created which corresponds to stock prices ranging from \$70-\$130.

Profit and Loss for both these legs (Long Stock and Short Call) are calculated against these spot prices.

- Long Stock: Profit/Loss = Spot - \$100 (or any other price at which the stock was bought).
- Short Call: Profit/Loss = Premium – max(Spot-Strike,0)

The results are plotted on a graph using matplotlib, which are as follows



On the right of the strike price is the capped profit zone. Breakeven point is at \$97 for the covered call. Long Stock follows the underlying completely, short call has capped profit on left of the strike + premium, after that it turns into loss. Covered call is the summation of these two legs and is also divided into 2 regions. On the right of the strike the profit is capped, on left of it has downside risk profile as shown in the figure.